

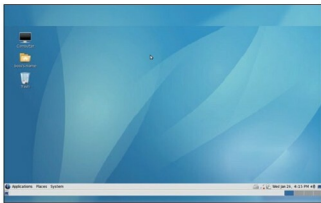
TECHNOLOGY AT YOUR SERVICE

Indian government to launch its own operating system, BOSS

Indian government is ready to unveil a new and improved version of Bharat Operating System Solutions (BOSS), developed by Centre for Development of Advanced Computing (C-DAC). This OS will replace Microsoft Windows and all other OSes in government offices in future.

BOSS is a Linux distribution, and is developed with the help of Gujarat Technical University, DRDO and some other private computer manufacturers. Initially developed in 2007 by National Resource Centre for Free/Open Source Software (NRCFOSS) of India, this OS is free and open source.

The new version has been tested successfully, defending itself from all kinds of attacks during the



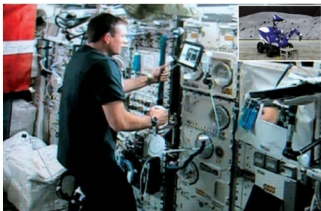
BOSS, Indian government's own operating system

past three months of trial. Several government agencies including the Army intelligence were given the task to attack it to test its vulnerability status but they all failed to break it. The new OS is believed to have been launched to overcome the vulnerabilities that exist in government cyberspace, resulting in a multitude of attacks by hackers.

The OS will be handed over to Department of Electronics and Information Technology (DeitY) to further smooth its rough edges and get it ready for use in government ministries.

Astronaut controls robot from space

Danish astronaut Andreas Mogensen has successfully controlled a robot on Earth with micrometre precision. He accomplished this feat while being 400km up and travelling at speeds of 28,800kmph. He used a feedback system to feel what the robot was doing, commanded Interact Centaur rover in real-time using a joystick to drive it around before taking control of its robotic arm to place a peg into a hole on a circuit board.



Danish astronaut Andreas Mogensen controlling the robot from aboard the International Space Station

This experiment was conducted to replicate connecting a wire on another planet. With a tolerance of only 150µm, it required precision and delicacy from both robot and human operator. The first attempt took 45 minutes, but the second took only ten minutes. In this attempt, Mogensen could touch the peg and sense how it felt using force-reflective technology in the joystick. Video streams captured by the robot gave him an overall view of his surroundings.

3D-printed microbots deliver drugs inside human body

Nano-engineers at University of California, San Diego, USA, have used an innovative 3D-printing technology to create tiny, multipurpose microbots that look like fish and are magnetically controlled. These bots are expected to have very diverse applications ranging from detoxification to drug delivery, among others.

The traditional bots were not able to perform complex tasks because of their simple structures, which were either cylindrical or spherical. The new technique removes this limitation, allowing researchers to create more complex and sophisticated geometrical structures.

The team, led by Prof. Shaochen Chen and Prof. Joseph Wang, put platinum nano-particles in the tail of the fish to react with hydrogen peroxide to propel the fish in forward direction. Magnetic iron-oxide particles in the head of microbots allowed the researchers to control the direction of these bots using magnets.

The team aims to build microbots that will perform surgery inside the human body with more accuracy and precision than currently possible.

Scientists develop stretchable metal

Scientists at Washington State University, USA, have discovered a way of stretching metal circuitry to twice